Yuankun Huang

Software Engineer — Game Systems & Simulation
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Game-systems engineer with **5+ years** shipping latency- and determinism-critical client systems for a live product with **10M+ downloads**. After hours I work on the Unity native boundary and on the core problems of simulation and validation: **deterministic record/replay**, clock synchronization, multi-agent simulation, and zero-allocation hot paths, all verified by **CI-enforced contracts**, not claims.

CORE SKILLS

Languages	C# (expert), C++20, Python; object-oriented design
Simulation & Validation	Deterministic record & replay, headless test gates, CI-enforced contracts, multi-agent simulation
Distributed & Runtime	.NET 8, SignalR, MessagePack, client networking (Protobuf), Unity (production, shipped)
Performance	Zero-allocation hot paths, object pooling, struct-oriented ECS, spatial partitioning, GPU instancing
Build & Quality	CMake, vcpkg, doctest, GitHub Actions CI, custom allocation guards
Certification	Unity Certified Expert: Programmer

PROFESSIONAL EXPERIENCE

Unity Developer May 2025 – Present
IGG Canada — Live Service & New-Title Client Development *Vancouver, BC*

- On an unannounced strategy title, owns the **live-event client framework** other engineers build on: data-layer base classes and registries, dependency-tree scheduling, and a content factory powering **20+ event types**; mentor new engineers on the team.
- Primary client owner of the **Showcase** and **Expedition** systems; researched the official **HybridCLR** integration path for the client hot-update pipeline.
- Rejoined **Mythic Heroes** live ops and delivered the 2025 live-event and monetization wave (**25 UI windows, 270+ C# files**) via monthly **ILRuntime** hot-updates; built **Analyze+**, an Addressables duplicate-reference analyzer.

Career Break: Bereavement & Family Care *Feb 2025 – May 2025*

Software Engineer (Contract, Hybrid) Jan 2024 – Jan 2025
The HIVE, UBC Faculty of Medicine — Gamified Rehabilitation Platform *Vancouver, BC*

- **Sole engineer** for TrAIT, a clinically validated Unity rehabilitation system (Leap Motion hand-tracking) with adaptive difficulty (Challenge Point Framework); owned all engineering, requirements through deployment.
- System achieved ∼**1,000 movements/session** vs. 32 in traditional therapy, supporting published results showing ∼**13% functional improvement** (*Neurorehabilitation and Neural Repair*).

Unity Developer Sep 2020 – Sep 2024
IGG Canada — Mythic Heroes (10M+ downloads) *Vancouver, BC*

- Core client engineer from early development to **global launch**; primary client owner of **Chat** and top-2 contributor on **Guilds**; shipped monthly content via **ILRuntime** hot-updates throughout live ops.
- Stood up the client network module for **Dawn of Survival**, a second shipped title: WebSocket socket/message managers, **Protobuf** serialization, gzip HTTP session layer, and pluggable login adapters.
- Profiling-driven optimization for low-end devices (Unity Profiler, Frame Debugger; AssetBundle restructuring) as part of a team push that cut load times ∼**30%**, build size ∼**20%**, and held stable **60 FPS** on mid-range Android for launch.

SELECTED PROJECTS

Rhythm Fruit Shop — Deterministic Native C++ Game Core [GitHub] *C++20 / SFML / miniaudio / CMake / CI*

- Implemented **deterministic record & replay** on a headless session core, validated in CI against an **independent Q16 fixed-point oracle**; **zero-allocation hot path** (custom `operator new` AllocationGuard) and layering enforced via **CMake dependency guards**.
- Built a real-time rhythm core where song time derives from the **audio backend's PCM cursor** through an **EMA-smoothed clock** (not frame deltas); input events carry host-monotonic timestamps reverse-mapped to song time for judgment-grade accuracy.

Kodama — Server-Authoritative Multi-Agent Simulation [GitHub] *.NET 8 / SignalR / MessagePack / Unity*

- Built a distributed real-time simulation sustaining **10K agents** with sub-millisecond server ticks: struct-oriented ECS, spatial partitioning, zero-GC hot path, and a state-interpolating renderer with GPU instancing; binary sync cut payloads ∼**70%** vs. JSON.

Synapse — Cross-Language Distributed Simulation [GitHub] *.NET 8 / Python / Unity*

- Streamed **2,000+ Python-driven agents** into a Unity client at stable **60 FPS**, using lock-free multi-threaded concurrency, pooling, and state conflation to hold per-update managed allocations near **1.4 KB**.

EDUCATION

M.S. Computer Science (Computing Systems), Georgia Institute of Technology (OMSCS, part-time online) Starting Fall 2026
Master of Digital Media, Centre for Digital Media, Vancouver 2021
B.Eng. Telecommunication Engineering, BUPT / Queen Mary University of London 2019